Project: Worked examples
Title: Enclosure heat dissipation and fan sizing
Status: DRAFT · Author: TempestOS (2026-10-23) · Checker: — (—) · Issued: —

## 1. Scope and method

Total heat load inside a control enclosure, the effective surface that can dissipate it for the mounting condition, the internal temperature rise with natural convection and radiation through a stated overall heat-transfer coefficient, the forced-ventilation airflow for a target rise (density corrected for altitude), the fan duty point against a fan curve and the system resistance, optional solar gain, and the cooling-unit capacity needed when the limit cannot be met.

| WHAT IT CALCULATES | RELATION |
|---|---|
| Heat dissipated inside the enclosure, plus any solar gain | $P = \Sigma P_i n_i d_i (+ \alpha I A_{sun})$ |
| Effective cooling surface, each face weighted by how freely it sheds heat | $A_{eff} = \Sigma f_{face} \times A_{face}$ |
| Temperature rise with natural cooling, one coefficient over the exposed faces (the IEC TR 60890 tables are not used) | $\Delta T = P/(h A_{eff})$ |
| Air flow a fan must move to hold the target rise | $V = P/(\rho c_p \Delta T_{target})$, with $\rho = p(z)/(287.055 T)$ and $c_p$ = 1006 J/(kg·K) |
| Fan duty point, where the fan curve meets the system resistance (found by bisection) | $p_{fan}(V) = p_{sys} (V/V_{ref})^2$ |
| Cooling unit capacity, when the surfaces alone cannot reject the heat | $P_{cool} = P - h A_{eff} (T_{max} - T_{amb})$ |
| Sealed enclosure (sealing temperature $T_{seal}$ entered, no breather), pressure difference across the walls when the air warms or cools from the sealing temperature at constant volume, in absolute temperatures (a rise when $T_{int} > T_{seal}$, a suction when below) | $\Delta p = p_{site} \cdot \lvert T_{int}/T_{seal} - 1 \rvert$ |
| A pressure-compensation element (breather or relief) limits it to its cracking pressure | $dp_{seal} = \min(\Delta p, dp_{relief})$ |
| Force of that pressure on a cover of area $A_{cover}$ | $F_{seal} = dp_{seal} \cdot A_{cover}$ |

Basis: A method of temperature-rise verification of low-voltage switchgear and controlgear assemblies by calculation[1]; Effective surface, heat-transfer coefficient and fan-sizing practice (cited, not copied)[2]; Verification of temperature rise[3](10.10).

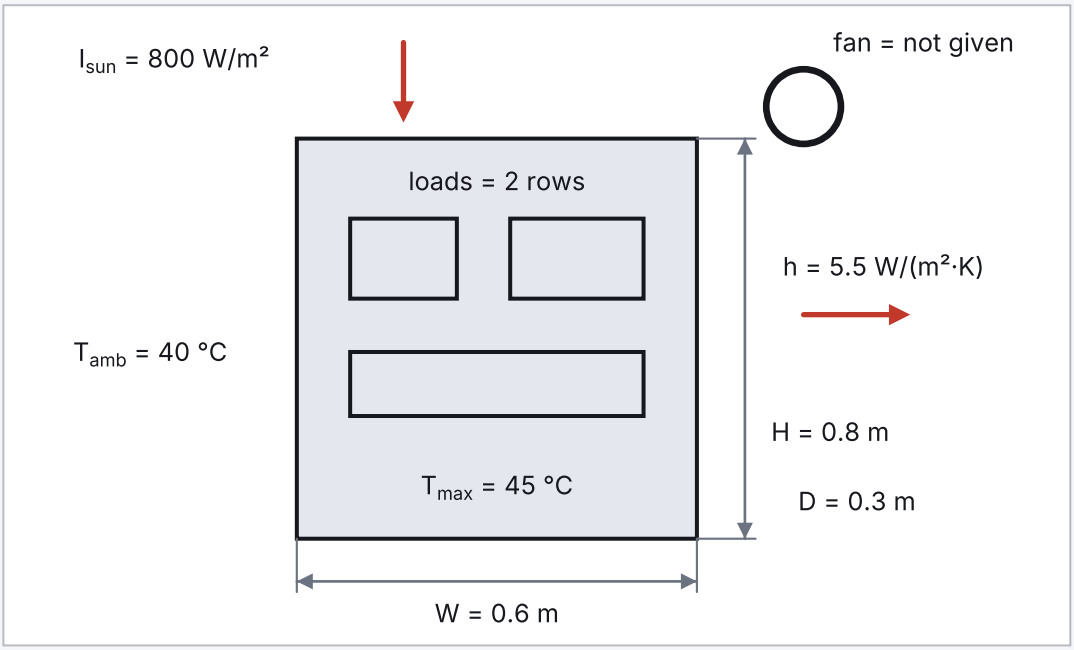



Figure 1 — An enclosure W by H by D holding the heat loads as listed, losing heat through its exposed faces to the ambient, with a fan and the sun on it. Not to scale; the labels are the inputs as entered.

## 2. Inputs

| SYMBOL | DESCRIPTION | VALUE | UNIT | SOURCE | VALID RANGE |
|---|---|---|---|---|---|
| loads | Heat loads | 2 row(s) | | spec: Panel schedule PNL-09 | any |
| $P_{add}$ | Additional heat load (linked from E6's dissipated heat or another calculation) | 0 | W | default | 0 … 100000 W |
| $I_{sun}$ | Solar irradiance on the enclosure (0 = indoors) | 800 | W/m² | spec: Site conditions SC-5: 800 W/m² on the sunlit side | 0 … 1200 W/m² |
| $A_{sun}$ | Sunlit area | 0.48 | m² | assumption: One 600 × 800 face in the sun. | 0 … 100 m² |
| $\alpha_{sun}$ | Surface absorptivity | 0.5 | | assumption: Light-grey paint. | 0.1 … 1 |
| W | Enclosure width | 0.6 | m | spec: Enclosure datasheet AE-1380 | 0.1 … 5 m |
| H | Enclosure height | 0.8 | m | spec: Enclosure datasheet AE-1380 | 0.1 … 5 m |
| D | Enclosure depth | 0.3 | m | spec: Enclosure datasheet AE-1380 | 0.1 … 5 m |
| $f_{top}$ | Top exposure (1 free, 0 covered) | 1 | | assumption: Free-standing kiosk, bottom on a plinth. | 0 … 1 |

| SYMBOL | DESCRIPTION | VALUE | UNIT | SOURCE | VALID RANGE |
|---|---|---|---|---|---|
| $f_{front}$ | Front exposure | 1 | | assumption: Free-standing kiosk, bottom on a plinth. | 0 … 1 |
| $f_{rear}$ | Rear exposure (0 = wall-mounted) | 1 | | assumption: Free-standing. | 0 … 1 |
| $f_{left}$ | Left side exposure | 1 | | assumption: Free-standing kiosk, bottom on a plinth. | 0 … 1 |
| $f_{right}$ | Right side exposure | 1 | | assumption: Free-standing kiosk, bottom on a plinth. | 0 … 1 |
| $f_{bottom}$ | Bottom exposure (0 = on the floor) | 0 | | assumption: Free-standing kiosk, bottom on a plinth. | 0 … 1 |
| h | Overall heat-transfer coefficient of the enclosure surface (editable reference value) | 5.5 | W/(m²·K) | reference: Reference record: painted steel enclosure, still air, 5.5 W/m²K | 3 … 8 W/(m²·K) |
| $T_{amb}$ | Ambient temperature | 40 | °C | spec: Site conditions SC-5 | -20 … 55 °C |
| $T_{max}$ | Maximum internal temperature | 45 | °C | spec: Site conditions SC-5 | 20 … 80 °C |
| altitude | Site altitude (air density) | 0 | m | assumption: Free-standing kiosk, bottom on a plinth. | 0 … 4000 m |
| $p_{sys}$ | System resistance (filters, grilles) at the reference airflow | 20 | Pa | default | 0 … 500 Pa |
| $V_{ref}$ | Reference airflow for the system resistance | 100 | m³/h | default | 1 … 100000 m³/h |

| TABLE ROW | VALUES |
|---|---|
| loads[1] | name = PLC and I/O; P = 80 W; qty = 1; duty = 1 |
| loads[2] | name = drives; P = 55 W; qty = 4; duty = 1 |

## 3. Assumptions

A1. Heat leaves the enclosure through its exposed faces at one overall coefficient h (natural convection plus radiation to surroundings at the ambient temperature); covered faces dissipate nothing. (IEC TR 60890-style effective-surface approach; h is a reference value the user may edit.)
A2. Forced ventilation removes the full heat load with air entering at ambient and leaving at the target rise; natural dissipation is neglected for the airflow (conservative). (Manufacturer fan-sizing practice.)
A3. A sealed enclosure ($T_{seal}$ entered) is a closed, rigid, leak-free volume of dry air at the site pressure when closed: the pressure difference follows from the ideal-gas relation at constant volume, from the sealing temperature to the internal temperature with natural dissipation $T_{int}$ (the temperature without a cooling unit); a breather, a seal that leaks or a flexing wall reduces it, so it is an upper bound (a

pressure-compensation element is modelled only as a cap at its cracking pressure, with no flow resistance); the site pressure is the standard atmosphere at the altitude. (Ideal gas at constant volume (derived).)
A4. The fan curve is linear between the entered points; the system resistance scales with airflow squared. (Duty-point idealisation.)
A5. $A_{sun}$ (Sunlit area) is taken as 0.48 m^2. (One 600 × 800 face in the sun.)
A6. $\alpha_{sun}$ (Surface absorptivity) is taken as 0.5. (Light-grey paint.)
A7. $f_{top}$ (Top exposure (1 free, 0 covered)) is taken as 1. (Free-standing kiosk, bottom on a plinth.)
A8. $f_{front}$ (Front exposure) is taken as 1. (Free-standing kiosk, bottom on a plinth.)
A9. $f_{rear}$ (Rear exposure (0 = wall-mounted)) is taken as 1. (Free-standing.)
A10. $f_{left}$ (Left side exposure) is taken as 1. (Free-standing kiosk, bottom on a plinth.)
A11. $f_{right}$ (Right side exposure) is taken as 1. (Free-standing kiosk, bottom on a plinth.)
A12. $f_{bottom}$ (Bottom exposure (0 = on the floor)) is taken as 0. (Free-standing kiosk, bottom on a plinth.)
A13. altitude (Site altitude (air density)) is taken as 0 m. (Free-standing kiosk, bottom on a plinth.)

## 4. Working

[18] $P_{loads} = \Sigma P_i \times n_i \times d_i$
$= 2$ item(s)
$= 300$ W
Internal heat load
[19] $P_{solar} = \alpha_{sun} * I_{sun} * A_{sun}$
$= 0.5 \times 800$ W/m² $\times 0.48$ m²
$= 192$ W
Solar gain
[20] $P_{total} = P_{loads} + P_{add} + P_{solar}$
$= 300$ W $+ 0$ W $+ 192$ W
$= 492$ W
Total heat to remove
[21] $A_{eff} = f_{top} * W * D + f_{bottom} * W * D + f_{front} * W * H + f_{rear} * W * H + f_{left} * H * D + f_{right} * D$ [1](approach)
$= 1 \times 0.6$ m $\times 0.3$ m $+ 0 \times 0.6$ m $\times 0.3$ m $+ 1 \times 0.6$ m $\times 0.8$ m $+ 1 \times 0.6$ m $\times 0.8$ m $+ 1 \times 0.8$ m $\times 0.8$ m $\times 0.3$ m
$= 1.62$ m²
Effective dissipating surface
[22] $dT_{nat} = P_{total} / (h * A_{eff})$
$= 492$ W $/ (5.5$ W/(m²·K) $\times 1.62$ m²)
$= 55.219$ K
Temperature rise with natural dissipation
[23] $T_{int} = T_{amb} + dT_{nat}$
$= 40$ degC $+ 55.219$ K
$= 95.219$ degC
Internal temperature with natural dissipation
[24] $dT_{allow} = T_{max} - T_{amb}$
$= 45$ degC $- 40$ degC
$= 5$ K
Allowed rise
[25] $u_{nat} = dT_{nat} / dT_{allow}$
$= 55.219$ K $/ 5$ K
$= 11.044$
Natural dissipation utilisation
[26] $P_{nat,max} = h * A_{eff} * dT_{allow}$
$= 5.5$ W/(m²·K) $\times 1.62$ m² $\times 5$ K
$= 44.55$ W
Heat the enclosure dissipates naturally at the limit

[27] $P_{cool}$ = max(0 W, $P_{total}$ - $P_{nat,max}$)
= max(0 W, 492 W - 44.55 W)
= 447.45 W
Cooling-unit capacity needed if ventilation is not used
[28] $p_{site}$ = 101325 (1 - 2.25577e-5 z)^5.2559
= z = 0 m
= 101.32 kPa
Site pressure
[29] $\rho_{air}$ = $p_{site}$ / (287.055 J/(kg*K) * $T_{amb}$)
= 101.32 kPa / (287.055 J/(kg*K) × 40 degC)
= 1.1272 kg/m³
Air density at ambient and altitude
[30] $V_{req}$ = $P_{total}$ / ($\rho_{air}$ * 1006 J/(kg*K) * $dT_{allow}$)
= 492 W / (1.1272 kg/m³ × 1006 J/(kg*K) × 5 K)
= 312.39 m³/h
Airflow for forced ventilation at the allowed rise (natural dissipation neglected)
[31] $V_{req,net}$ = max(0 m^3/h, ($P_{total}$ - $P_{nat,max}$) / ($\rho_{air}$ * 1006 J/(kg*K) * $dT_{allow}$))
= max(0 m^3/h, (492 W - 44.55 W) / (1.1272 kg/m³ × 1006 J/(kg*K) × 5 K))
= 284.11 m³/h
Airflow crediting natural dissipation

## 5. Results

| SYMBOL | VALUE | DESCRIPTION |
|---|---|---|
| $P_{loads}$ | 300 W | Internal heat load |
| $P_{solar}$ | 192 W | Solar gain |
| $P_{total}$ | 492 W | Total heat |
| $A_{eff}$ | 1.62 m² | Effective surface |
| $dT_{nat}$ | 55.219 K | Natural temperature rise |
| $T_{int}$ | 95.219 degC | Internal temperature (natural) |
| $P_{nat,max}$ | 44.55 W | Natural dissipation at the limit |
| $P_{cool}$ | 447.45 W | Cooling unit needed |
| $T_{max}$ | 45 degC | Internal temperature limit |
| $\rho_{air}$ | 1.1272 kg/m³ | Air density |
| $V_{req}$ | 312.39 m³/h | Required airflow |
| $V_{req,net}$ | 284.11 m³/h | Required airflow (net) |

## 6. Checks (governing first)

| RESULT | CHECK | DEMAND | CAPACITY | UTILISATION | REF |
|---|---|---|---|---|---|
| FAIL | Natural dissipation holds $T_{max}$ | $u_{nat}$ = 11.044 | $u_{nat}$ ≤ 1 | 11 | |

## 7. Applicability

WARNING — input.P_add: $P_{add}$ (Additional heat load (linked from E6's dissipated heat or another calculation)) was not entered; the template default 0 W is used.
WARNING — input.p_sys: $p_{sys}$ (System resistance (filters, grilles) at the reference airflow) was not entered; the template default 20 Pa is used.
WARNING — input.V_ref: $V_{ref}$ (Reference airflow for the system resistance) was not entered; the template

default 100 m^3/h is used.
WARNING — cooling-unit: $T_{max}$: natural dissipation holds 44.55 W of 492 W; ventilate or fit a 447.4 W cooling unit.

## 8. Verification

```
Template elec.enclosure-heat@1.5.0 Content hash 857f14f33a5fb9c0 Engine calc-engine/1.0.0
Known-answer cases passed 4/4 Test run 2026-10-23 12:00 UTC TempestOS 0.26.0
```

## 9. References

1. International Electrotechnical Commission. A Method of Temperature-Rise Verification of Low-Voltage Switchgear and Controlgear Assemblies by Calculation. IEC TR 60890. IEC.
2. Rittal; nVent Hoffman. Enclosure climate control guides [manufacturer technical literature].
3. British Standards Institution. Low-Voltage Switchgear and Controlgear Assemblies—Part 1: General Rules. BS EN 61439-1. BSI.